

Experiment 18: DB with Name and DOB

1. Aim

To write and execute a Prolog program to create a database containing the **name and date of birth (DOB)** of persons and retrieve the required information using queries.

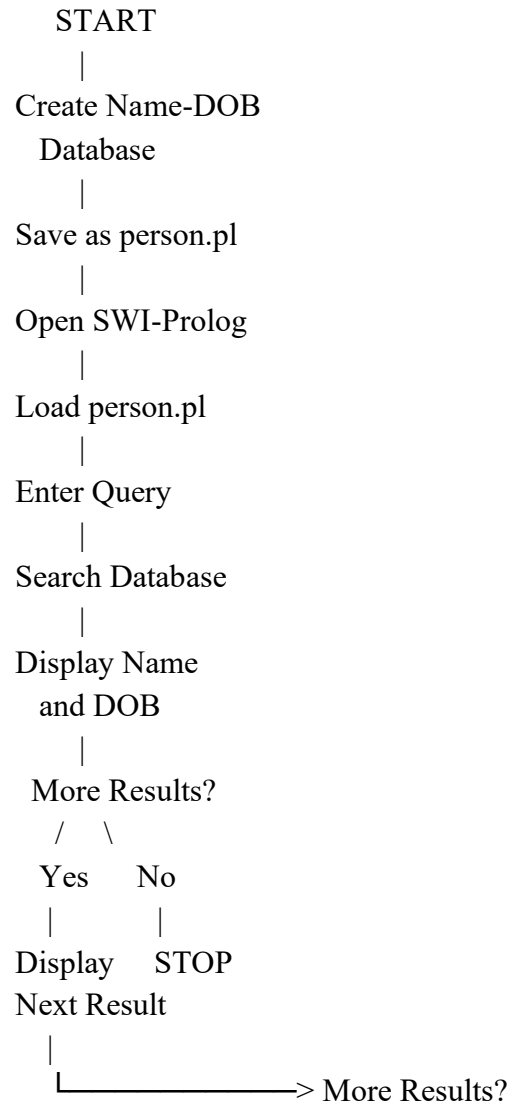
2. Objective

- To understand facts and queries in Prolog.
- To create a simple database using Prolog facts.
- To retrieve a person's date of birth using their name.
- To display all persons and their corresponding dates of birth.

3. Algorithm

1. Start.
2. Create Prolog facts containing the name and DOB of different persons.
3. Save the program as person.pl.
4. Open SWI-Prolog.
5. Load the program using the consult() command.
6. Enter a query to search for a person's DOB.
7. Display the matching result.
8. Use variables to retrieve all names and DOBs.
9. Stop.

4. Flowchart



5. Prolog Program

```
person(devadharshini,'16-09-2007').  
person(rahul,'10-05-2006').  
person(priya,'20-11-2005').  
person(arun,'12-03-2007').  
person(kavya,'25-07-2006').  
person(rohan,'08-01-2005').  
person(anitha,'14-12-2007').  
person(vijay,'30-06-2006').  
person(sneha,'22-02-2005').  
person(karthik,'05-10-2007').
```

display :-

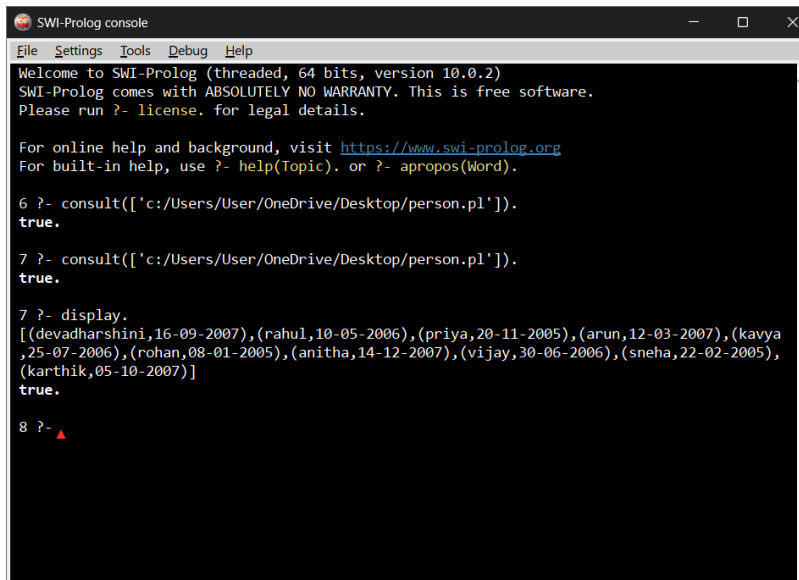
```
findall((X,DOB), person(X,DOB), List),  
write(List).
```

Query 1 – Display all persons

```
person(X,DOB).
```

Press ; after each result to display the next record.

6. Sample Output



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

6 ?- consult(['c:/Users/User/OneDrive/Desktop/person.pl']).
true.

7 ?- consult(['c:/Users/User/OneDrive/Desktop/person.pl']).
true.

8 ?- display.
[(devadharshini,16-09-2007),(rahul,10-05-2006),(priya,20-11-2005),(arun,12-03-2007),(kavya,25-07-2006),(rohan,08-01-2005),(anitha,14-12-2007),(vijay,30-06-2006),(sneha,22-02-2005),(karthik,05-10-2007)]
true.

9 ?-
```

7. Result

Thus, the Prolog program for creating a **Name and Date of Birth database** was successfully implemented and executed. The required person's DOB and all database records were retrieved successfully using Prolog queries.

8. Conclusion

The experiment demonstrates how Prolog can be used to represent information as **facts** and retrieve information using **queries and variables**. It provides a simple way to create and search a knowledge database.